

CSE 4111 - MACHINE LEARNING - GROUP PRESENTATION

Artificial Bee Colony



Finding the global minimum of the 2-D Ackley function

SUBMITTED TO

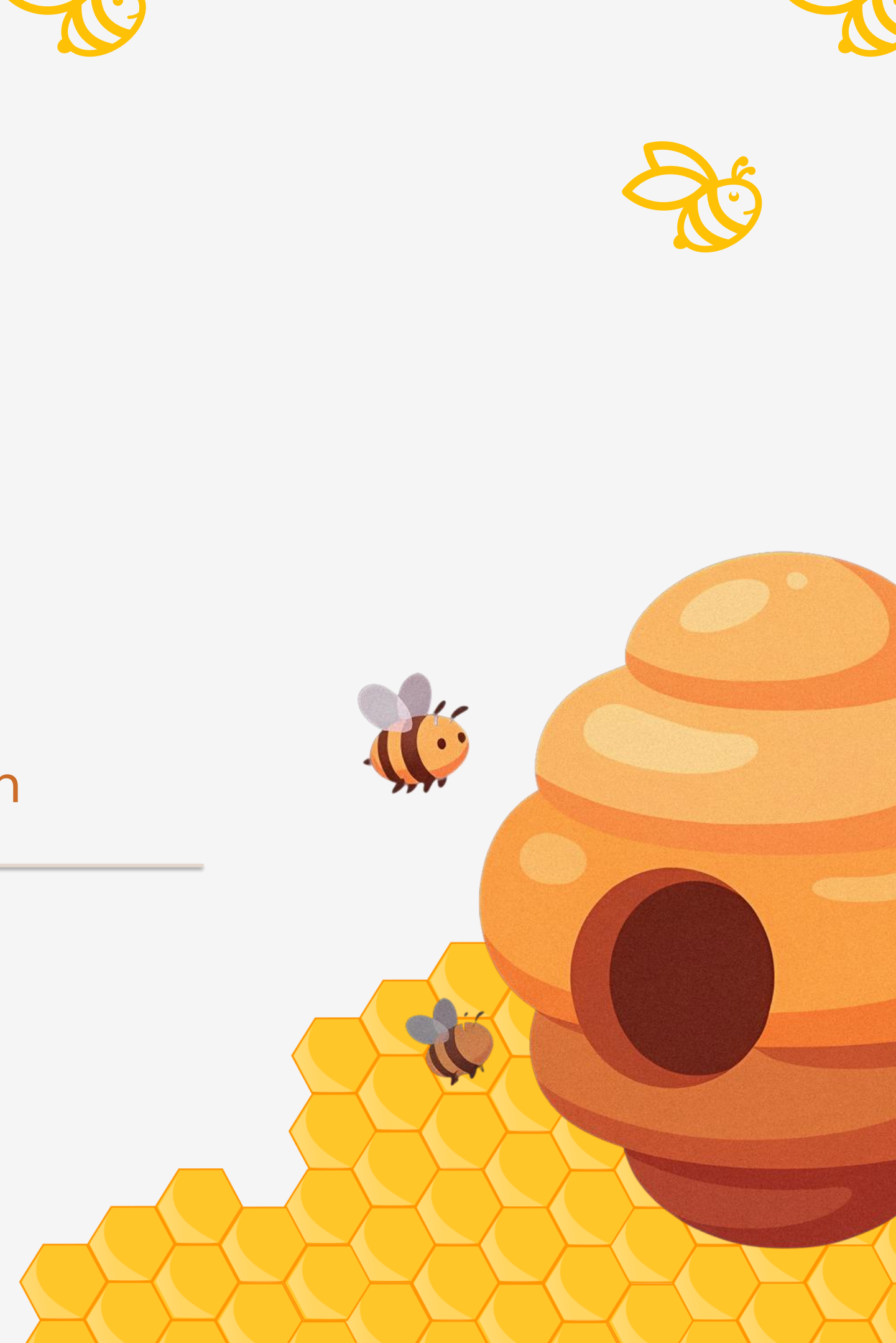
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Find the lowest point

01

Choose two numbers

Pick x_1 and x_2 so that f is as small as possible.

02

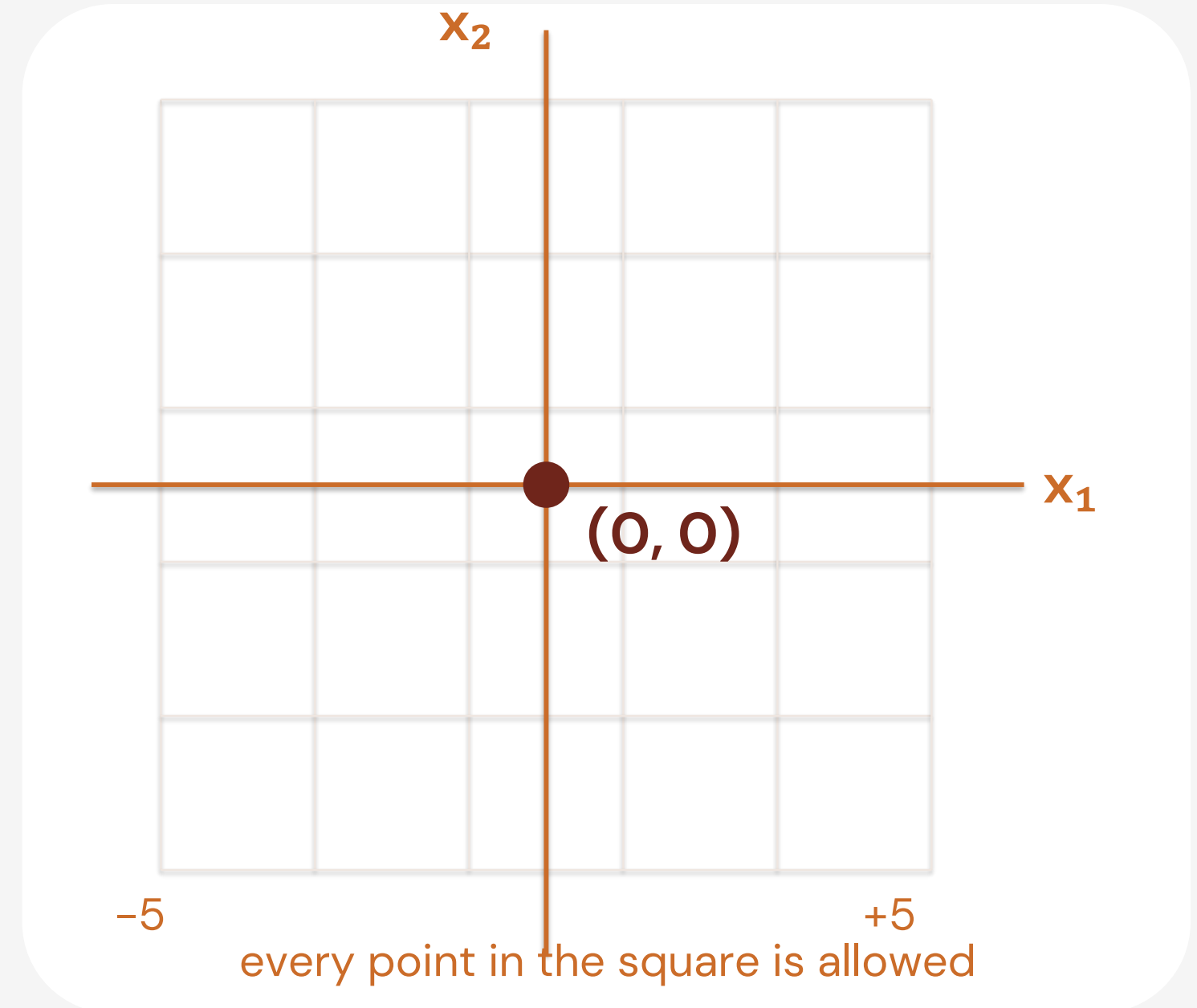
Stay inside the square

Both coordinates must satisfy $-5 \leq x \leq +5$.

03

The answer we are aiming at

$x^* = (0, 0)$ and $f(x^*) = 0$.



The bees only taste the points they choose. Nobody tells them where $(0, 0)$ is.

The Ackley function in 2-D

START WITH n VARIABLES

$$f(\mathbf{x}) = -a \exp\left(-b \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(cx_i)\right) + a + e$$

Set $n = 2$ 

Expand the two sums

Use $a = 20$, $b = 0.2$, $c = 2\pi$

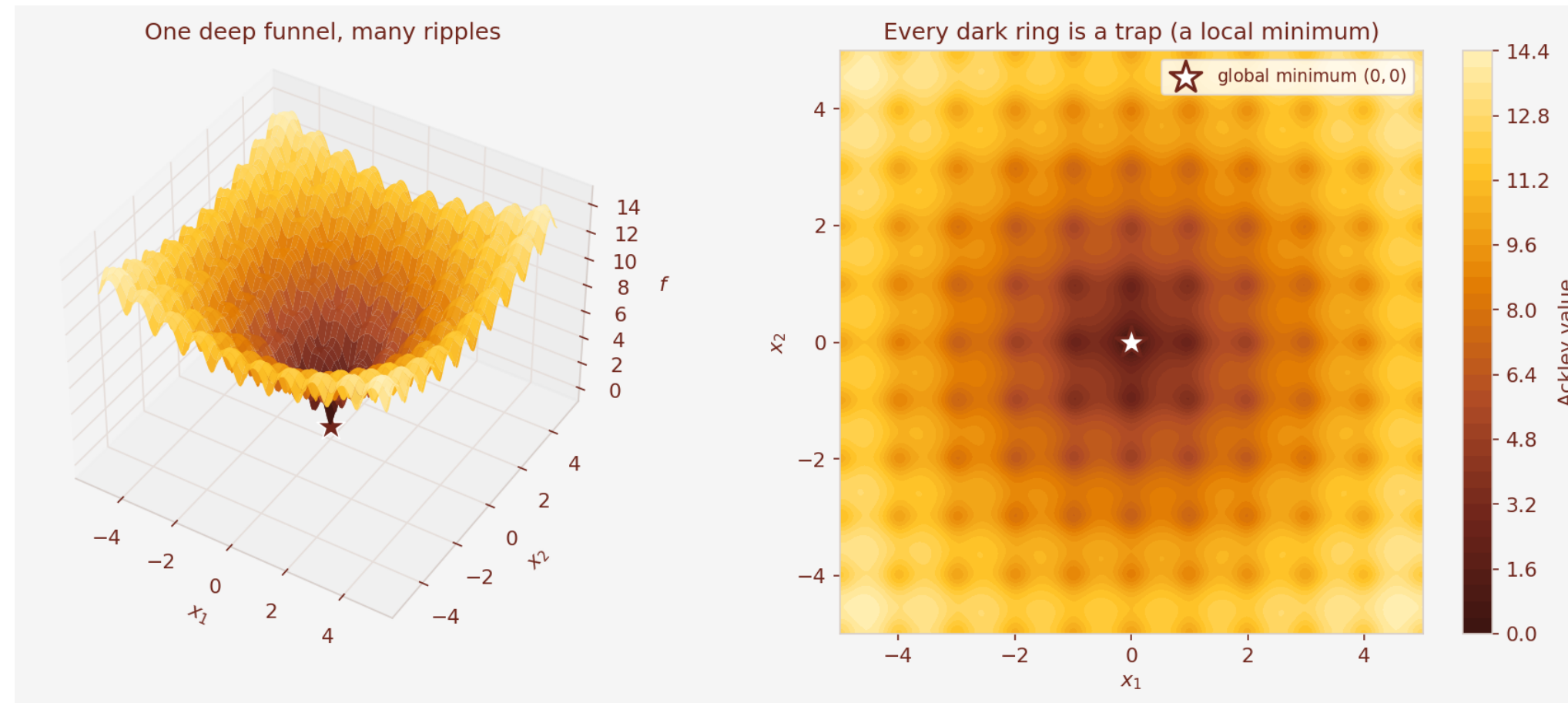
THE FUNCTION IN OUR CODE

$$f(x_1, x_2) = -20 \exp\left(-0.2 \sqrt{\frac{x_1^2 + x_2^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x_1) + \cos(2\pi x_2)}{2}\right) + 20 + e$$

Check it at the origin: $-20 - e + 20 + e = 0$



One deep minimum. Many small traps.



01

One wide funnel

The smooth part slopes towards the centre.

02

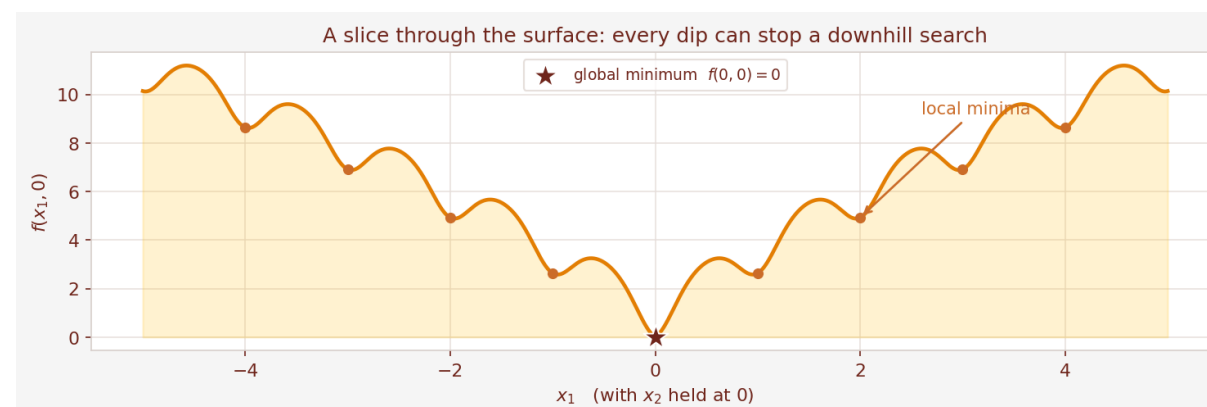
Many small traps

The cosine part adds dips everywhere.

03

Too big to scan

A 10^{-6} grid would need $\sim 10^{14}$ points.



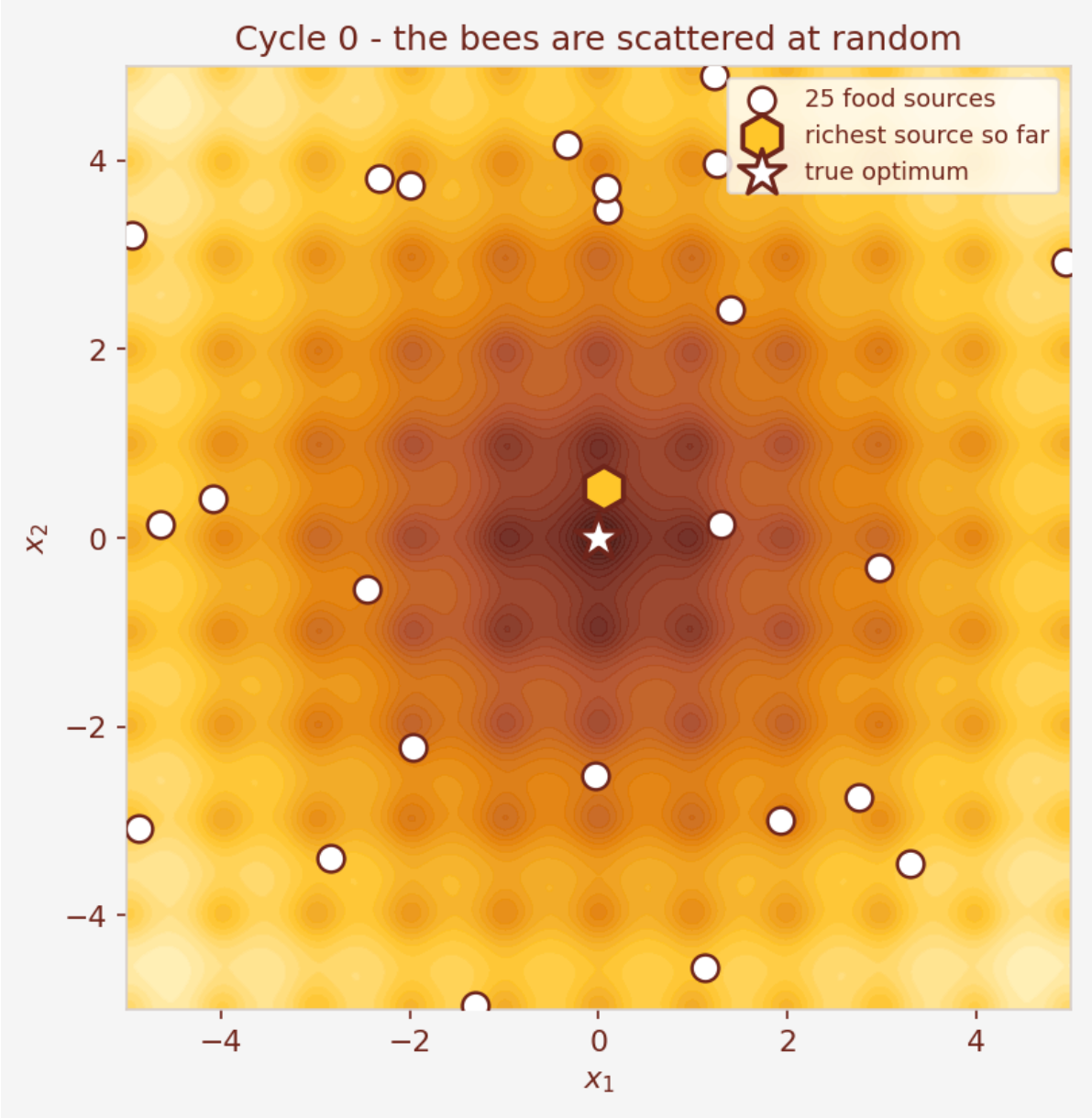
**A plain downhill search
can stop in any of the dips.**

A colony instead of a population

BEE LANGUAGE → OUR PROGRAM

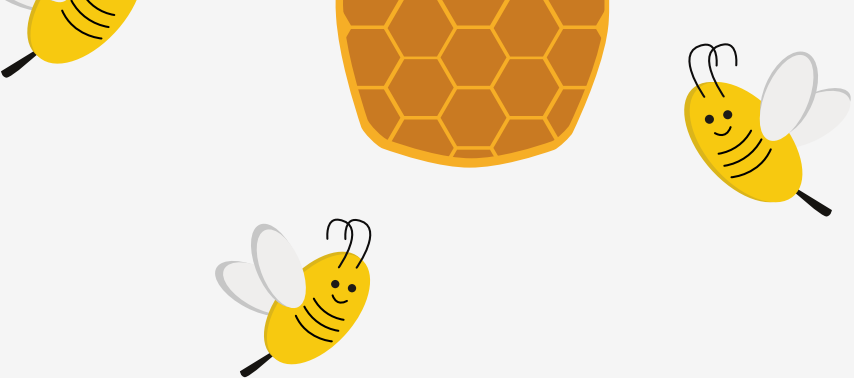
In the hive	In the program
Food source	one answer $[x_1, x_2]$
Nectar amount	fitness, bigger is better
Employed bee	improves its own source
Onlooker bee	helps the richest sources
Scout bee	restarts a stuck source

Published by Dervis Karaboga in 2005,
after the foraging behaviour of honey bees.

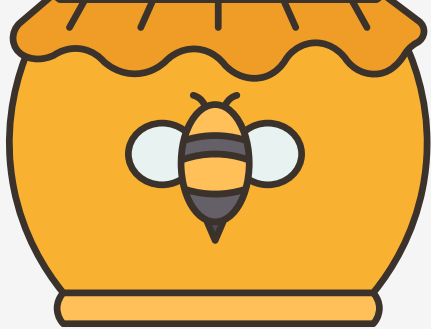


25 food sources explore at once. Better sources attract more of the colony's effort.

Three bees, three jobs



Exploitation, guided search and exploration — three jobs, repeated every cycle.



The settings we used

THE SIX CONTROL PARAMETERS (SLIDE 20)

Parameter	Value
Colony size	50 bees
Employed bees = sources (SN)	25
Onlooker bees	25
Dimension (D)	2
limit	50

limit = SN × D = 25 × 2 = 50
Seed 7, so the run repeats exactly.

THE SAME BUDGET, DIFFERENT MACHINERY

Genetic Algorithm	Artificial Bee Colony
Population 50	Colony 50 bees
100 generations	100 cycles
Crossover Pc = 80 %	not used
Mutation Pm = 5 %	not used
Mutation step sigma	free: x_i - x_k
Elitism = 1	memorise the best

Two blank rows: ABC has no crossover rate and no mutation rate to choose.



The ABC cycle

1 Assign the parameters

colony 50, SN 25, limit 50, 100 cycles

2 Initialise 25 sources

random points, evaluate, remember the best



4 Stop and report

after 100 cycles, or when progress stalls

↺ **back to step 3**

Step 3 – repeat every cycle

Employed bee phase

25 bees search next to their own source

Onlooker bee phase

25 roulette-wheel visits to the richer sources

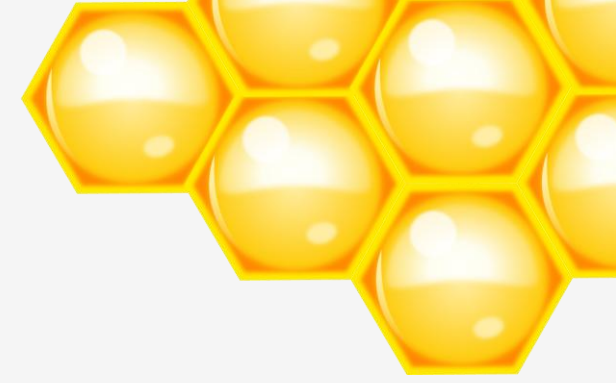
Memorise the best solution

done BEFORE the scouts run

Scout bee phase

replace any source stuck for 50 tries

Memorising the best before the scouts run is this algorithm's elitism.



Every solution is two real numbers

ONE FOOD SOURCE

COORDINATE 1

x_1

COORDINATE 2

x_2

both inside [-5 , +5]

INITIALISE — AND THE SCOUTS REUSE THE SAME LINE

$$X_{ij} = X_{\min,j} + \text{rand}(0, 1) (X_{\max,j} - X_{\min,j})$$

Example sources

[-4.0 , 2.5]

[-1.5 , -1.0]

Value encoding: the food source stores the two coordinates directly.



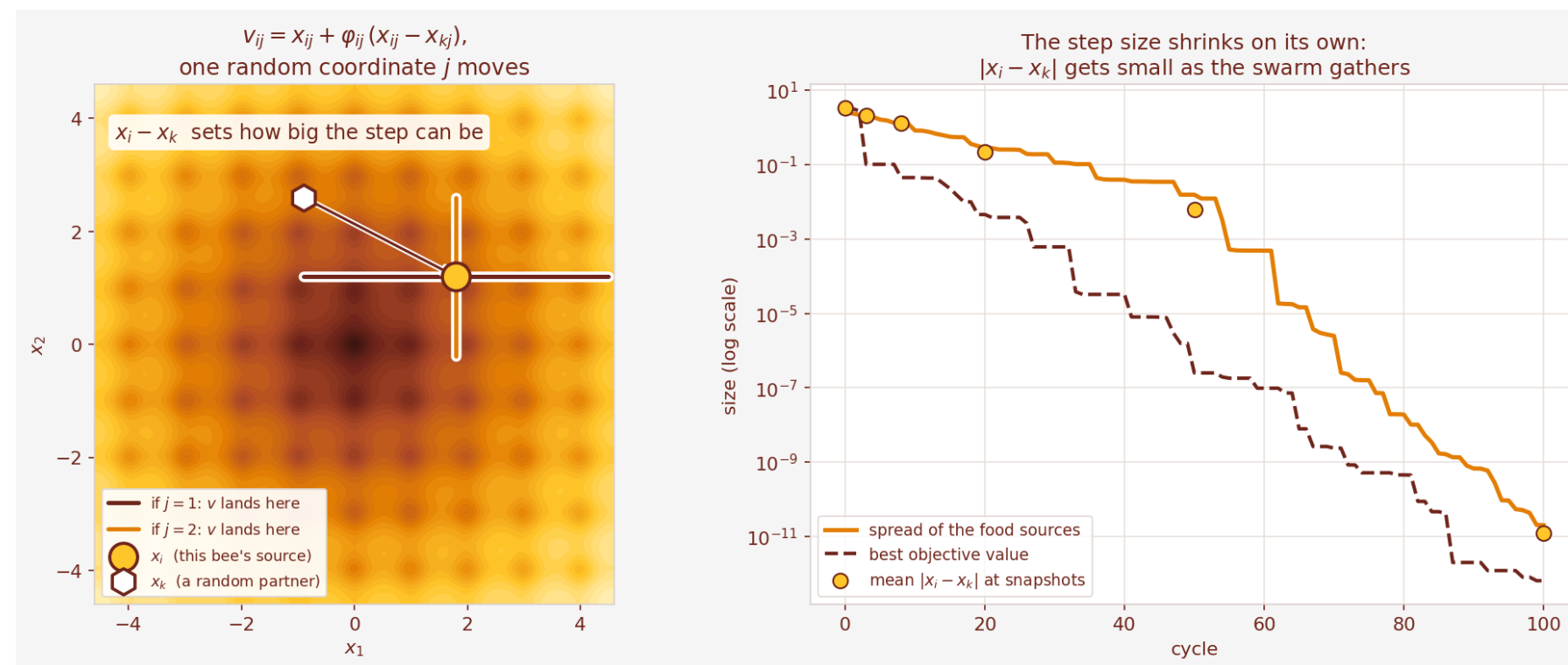
How a bee searches nearby

$$V_{ij} = X_{ij} + \varphi_{ij} (X_{ij} - X_{kj})$$

$\varphi = \text{rand}(-1, 1)$

$k \neq i$, another source

j = one random coordinate



The step is a fraction of the gap to another bee — so it shrinks by itself.

Keep the better one

old source x_i



trial point v_i

compare $f(x_i)$ with $f(v_i)$

v_i is better

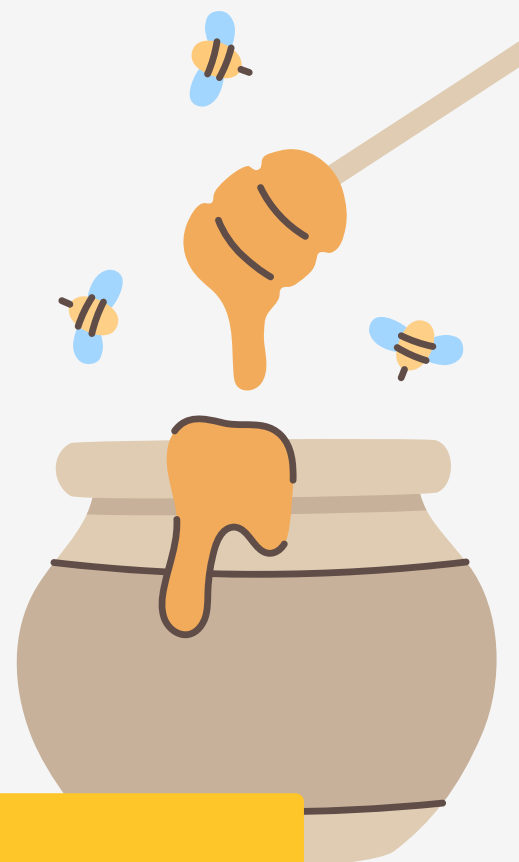
Replace the source with v_i .
Reset its trial counter to 0.

$\text{trial} \rightarrow 0$

v_i is worse

Keep the old source.
Add one to its trial counter.

$\text{trial} \rightarrow \text{trial} + 1$



Greedy selection: a food source can never get worse. The counter is what scouts watch.

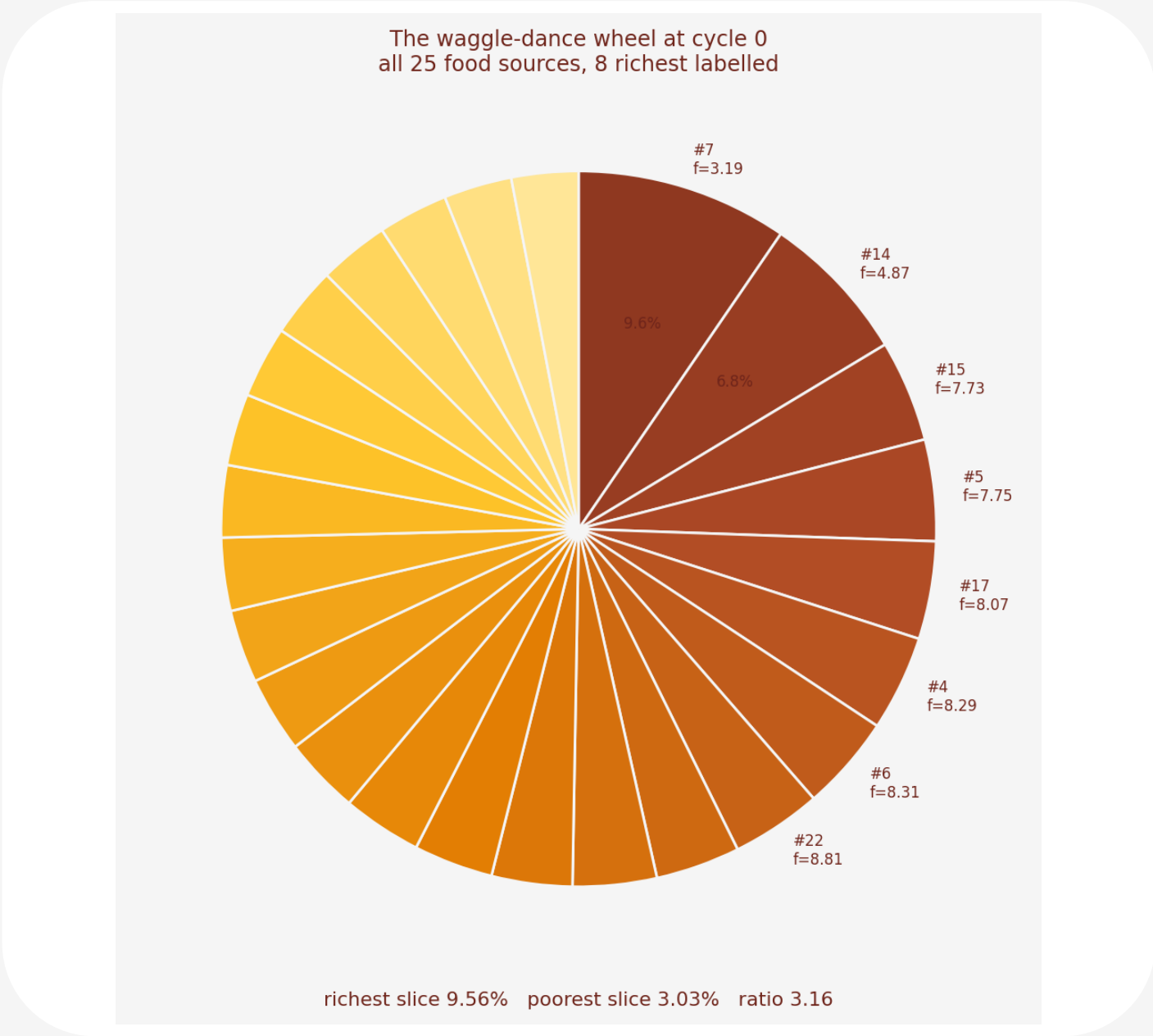
Better sources get larger slices

LOWER *f* → MORE NECTAR

$$fit_i = \frac{1}{1 + f_i} \quad (f_i \geq 0), \quad P_i = \frac{fit_i}{\sum_{m=1}^{25} fit_m}$$

Ackley is never negative, so the second branch of the standard rule is never used here.

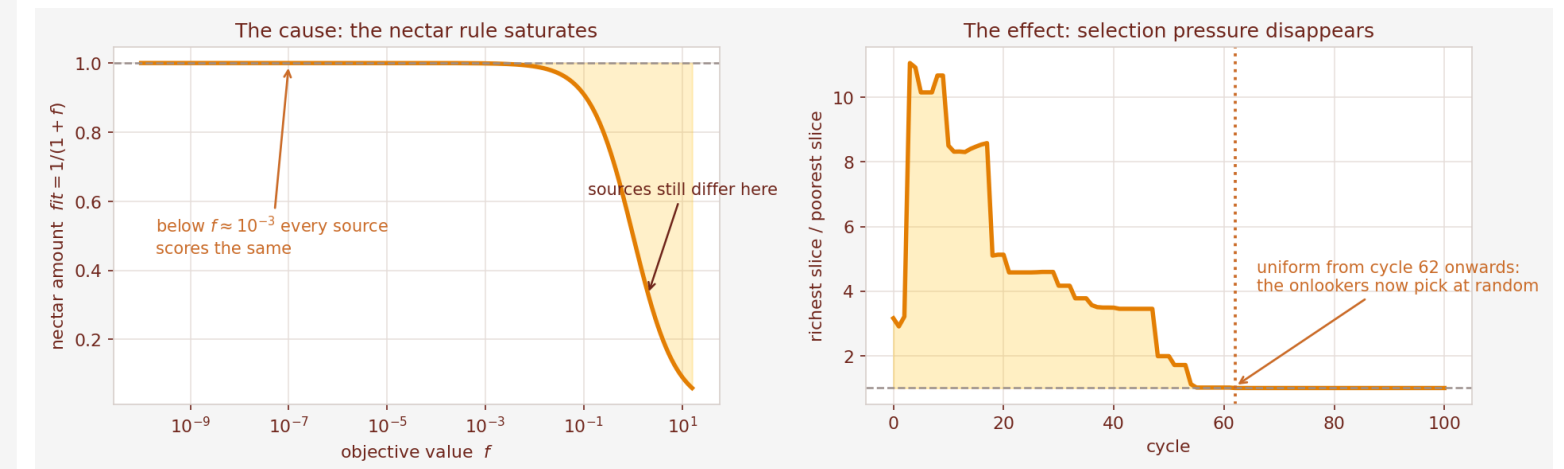
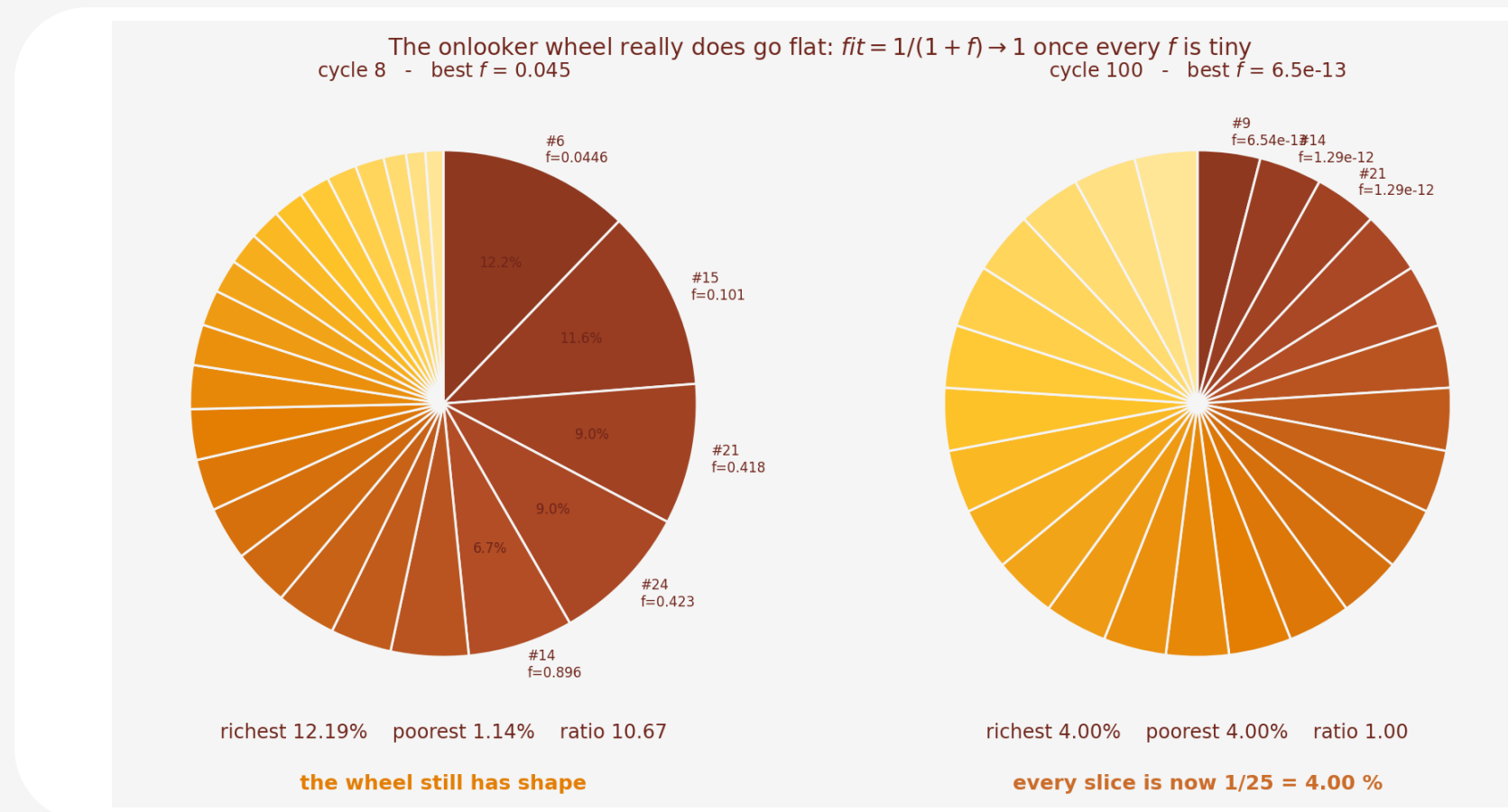
Source	<i>f</i>	Share of wheel
#7 (richest)	3.19	9.56 %
#14	4.87	6.82 %
#15	7.73	4.59 %
#3 (poorest)	12.23	3.03 %



Each of the 25 onlookers spins this wheel once. Even the poorest source keeps 3 %.



Something the slides do not mention



01

The nectar rule has a ceiling
 $fit = 1/(1+f)$ can never be larger than 1.

02

So the wheel goes flat
 From cycle 62 every slice is exactly 4.00 %.

We measured it instead of guessing. The next slide shows what it actually costs.

Which part of ABC actually matters?

Every row changes one thing and runs the same 40 seeds (0–39).

What we changed	median f	worst f	reached 10 ⁻⁶	scouts sent
full ABC as taught	5.476e-13	1.034e-11	40 / 40	0
no onlooker phase	2.999e-07	4.262e-06	36 / 40	0
no scout phase	5.476e-13	1.034e-11	40 / 40	0
perturb both coordinates	2.509e-13	4.267e-12	40 / 40	0
limit = 10	2.256e-06	1.071e-04	16 / 40	1,562
limit = 5	5.358e-04	1.248e-02	0 / 40	2,107
colony 10 (5 sources)	2.647e-04	4.121e-01	11 / 40	260

Onlookers do the precision work

Remove them and the median is 500,000x worse.

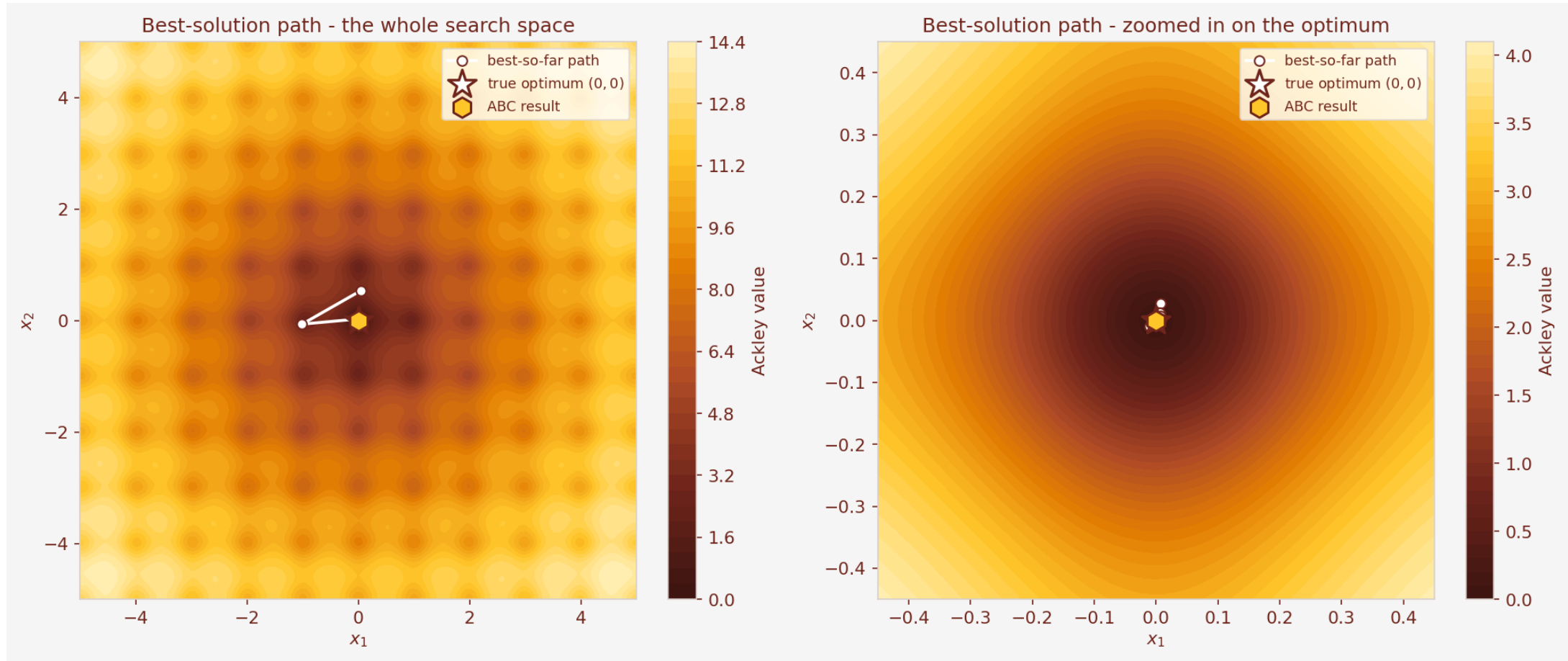
Scouts never fired here

Turning them off changed nothing at all.

A small limit is destructive

limit = 5 threw away good sources: 0 / 40.

Our result is very close to (0, 0)



6.541×10^{-13}

best Ackley value, seed 7

99 / 100

cycle the best was found

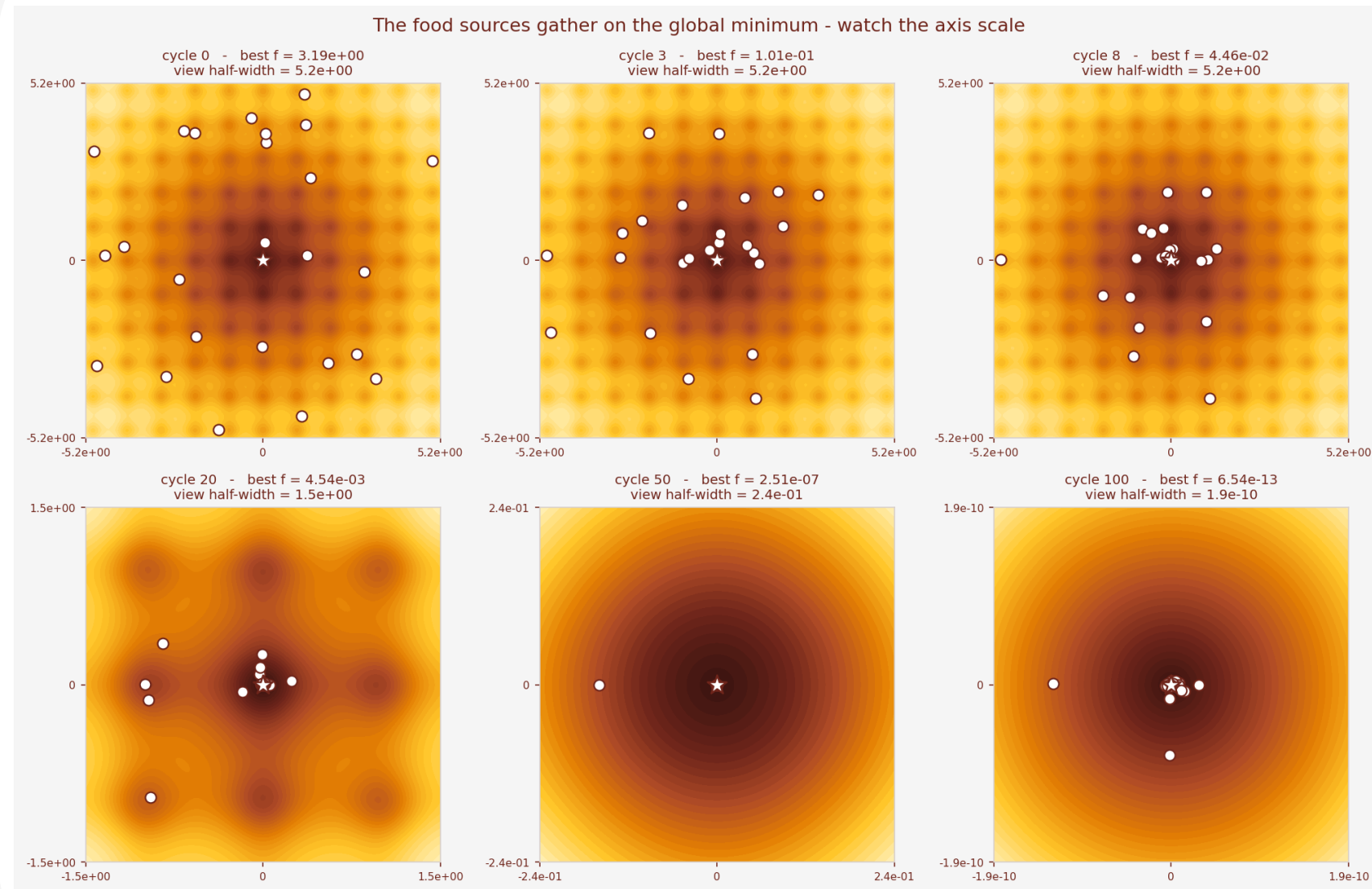
5,025

Ackley evaluations used, 0 scouts

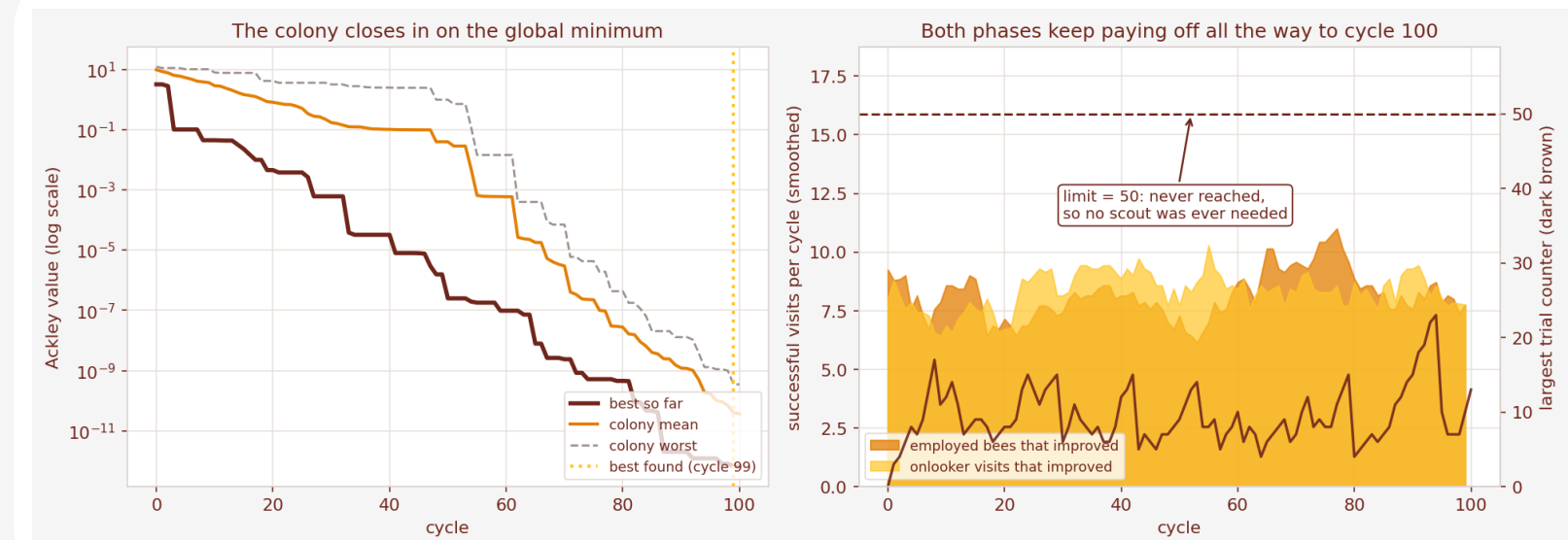
$x_1 = +2.278 \times 10^{-13}$ $x_2 = -3.938 \times 10^{-14}$ distance = 2.312×10^{-13}

random search: 0.769

The colony closing in



the colony contracts



The best value never rises

it is memorised at the end of every cycle

Both phases keep working

improvements continue right up to cycle 100

The trial counter peaked at 23

the limit of 50 was never reached, so no scout flew

Read the axis numbers on the snapshots: the last panel is only 2×10^{-10} wide.

All 40 runs met the target

40 / 40

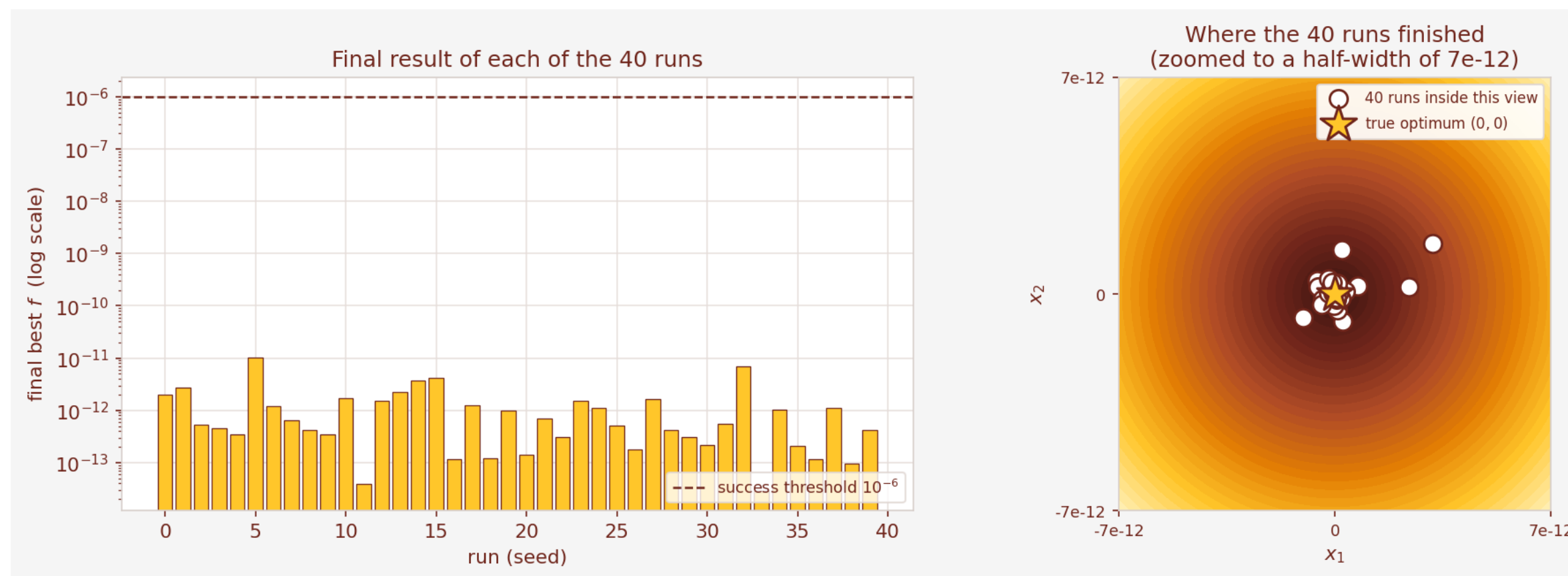
runs finished below 10^{-6}

5.476×10^{-13}

median final value

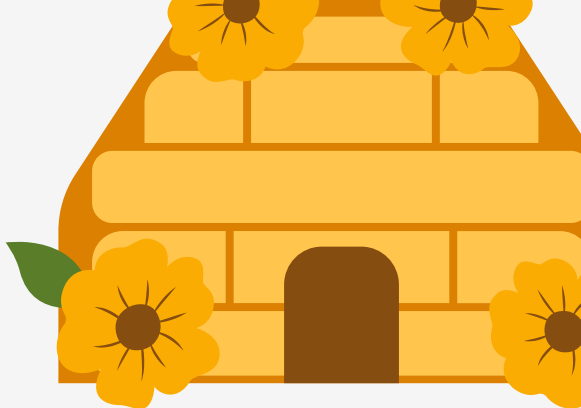
1.034×10^{-11}

worst final value



Evidence of reliability on this problem — not a guarantee for every possible run.

Bees next to genes

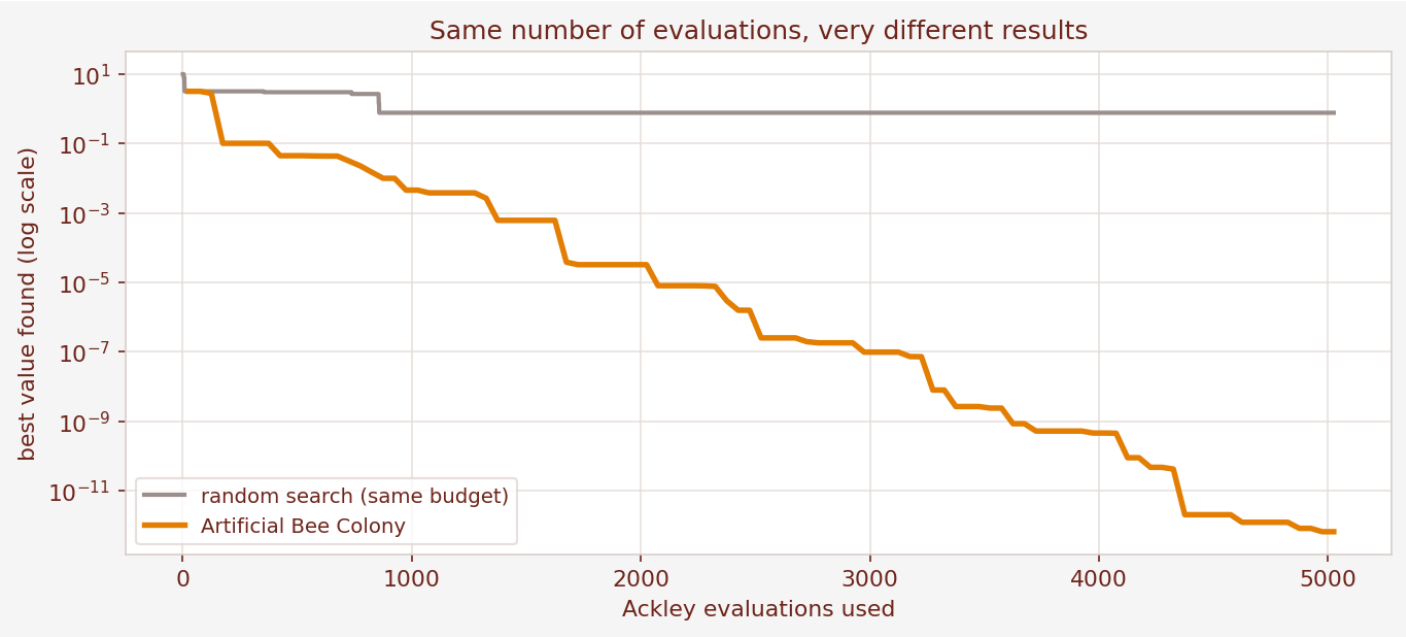


	Genetic Algorithm	Artificial Bee Colony
Ackley evaluations	5,050	5,025
Median of 40 runs	1.636e-12	5.476e-13
Worst of 40 runs	3.137e-11	1.034e-11
Runs below 1e-6	40 / 40	40 / 40
Parameters to tune	5	1

Same reliability, three times closer, one parameter to tune instead of five.

One function, two dimensions, 40 seeds. Evidence — not a claim about every problem.

The same evaluation budget, spent differently



The Genetic Algorithm needed a crossover rate, a mutation rate, a mutation-step rule, elitism and a duplicate cap. ABC needed one number: the limit.



What we learned



Three simple roles are enough

Repeated 100 times, they reach 13 decimal places.



The step size comes free

It is the distance between two bees, so it shrinks.



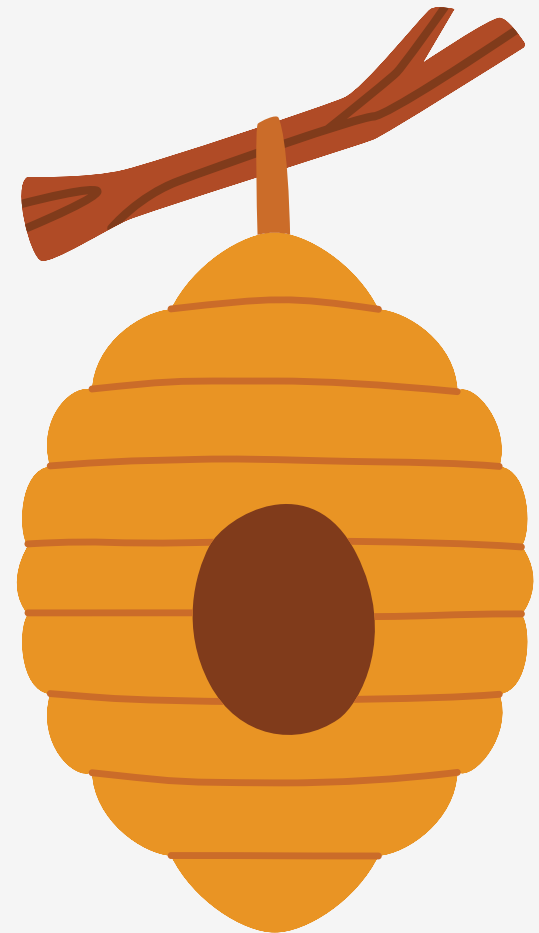
Test it, do not trust the diagram

The scout phase never ran once on this problem.



Say what you did not test

One function, two dimensions, 40 seeds.



Thank you. Questions?

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